Qihang Sun | Curriculum Vitae

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RESEARCH INTEREST

Economics of Transportation, Traffic Modeling, Optimization, Pedestrian Dynamics

EDUCATION

Beihang University **Beijing, CHN**
Ph.D. Student in Management Science and Engineering *2023.09 - Present*

Beihang University **Beijing, CHN**
B.S. in Biomedical Engineering (Minor) *2020.09 - 2023.06*

Beihang University **Beijing, CHN**
B.S. in Information Management and Information System *2019.09 - 2023.06*

AWARDS & HONORS

- **Outstanding Graduate Student**, Beihang University, 2023
- **Merit Student**, Beihang University, 2022
- **Model Student of Academic Records**, Beihang University, 2022
- **The Grand Price Academic Scholarship**, Beihang University, 2022
- **The First Prize Discipline Competition Scholarship**, Beihang University, 2022
- **The Second Prize Social Work Scholarship**, Beihang University, 2022
- **Honorable Mention Prize in Mathematical Contest In Modeling (MCM)**, COMAP, 2022
- **The Second Prize in China Undergraduate Mathematical Contest in Modeling**, CUMCM, 2021
- **The Second Price Academic Scholarship**, Beihang University, 2021
- **Outstanding Volunteer**, Beihang University, 2021

RESEARCH EXPERIENCE

Flexible traffic demand management strategy considering violations *2023.11 – Present*

- Designed a flexible traffic demand management strategy to reduce traffic congestion of morning commuting system.
- Introduced a scheme in consideration of violations to reduce the deadweight losses to the system.
- Derived the total travel cost under the equilibrium condition to prove the effectiveness of the strategy.

Alleviation of vertical congestion based on elevator retrofit *2023.09 – 2023.11*

- Developed a novel retrofit scheme of elevators which converted the passenger elevator into cargo elevator to alleviate vertical congestion in peak time.
- Demonstrated the theoretical and practical feasibility of the retrofit scheme.

Pedestrian flow modeling and analysis considering velocity heterogeneity *2022.10 – 2023.08*

- Constructed continuous social force model of pedestrian dynamics, to reveal the movement characteristics of pedestrians more accurately.
- Verified the validity of model construction and parameter setting by real pedestrian movement data.
- Investigated the dynamic evolution of the self-organized phenomenon.

Development of a novel medical data management system *2021.03 - 2021.06*

- Established a medical database management system which can realize ECG data collection, storage, query, EDF standardized input and output and ECG real-time rendering .
- Laid a foundation for the subsequent implementation of data analysis, data mining, deep learning and other functions

INTERNSHIP EXPERIENCE

Ministry of Industry and Information Technology *2022.03-2022.09*
Undergraduate Research Assistant

- Participated in the construction of national digital infrastructure, collected and obtained macroeconomic indicators, monitored the operation of the electronics industry, and redacted a complete digital economy industry analysis series report.
- Developed two information systems to ensure the efficient and orderly daily business of the department.

SKILLS

Programming	Python, R, Matlab, C, SQL, LaTeX
Languages	Chinese (*Native*), English (*Conversational*)
Software	MS Office, IBM SPSS
Sports	Football, Badminton, Table tennis

PUBLICATIONS

- **New insights into bi-directional pedestrian flows with heterogeneous speeds: modeling and analyses**
  Accepted by The 14th Workshop on Computational Transportation Science
  Sun Qihang, Guo Renyong
- **Effectiveness of deposit scheme and flexible reservation strategy in morning commute management**
  Working Paper
  Sun Qihang, Guo Renyong

CONFERENCE TALKS

- **The 14th Workshop on Computational Transportation Science**, Shanghai, 2023.08
  Oral Presentation

SELECTED COURSES

- **Ph.D. Student:**
  Advanced Microeconomics Theory (94), Advanced Mathematical Programming (93), Optimization Method (98)
- **Undergraduate:**
  Information Systems Analysis and Design (95), Program Designing (92), Production and Operations Management (94), Logistics and Supply Chain Management (96), Artificial Intelligence and Deep Learning (97), Decision-Making and Business Intelligence Systems (96), Large-Scale Database Application Experiment (96)

REFERENCES

- **Dr. Renyong Guo**
  Professor, School of Economics and Management
  Beihang University, Beijing, CHN
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